

## SAFETY DATA SHEET

Creation Date 13-April-2011

Revision Date 28-December-2021

Revision Number 4

### 1. Identification

**Product Name** N-epsilon-FMOC-alpha-BOC-L-Lysine

**Cat No. :** AC439080000; AC439080050

**CAS-No** 84624-27-1

**Synonyms** BOC-Lys(FMOC)-OH; N-alpha-BOC-N-epsilon-FMOC-L-Lysine

**Recommended Use** Laboratory chemicals.

**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

**Importer/Distributor**  
Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

Acros Organics  
One Reagent Lane  
Fair Lawn, NJ 07410

**Manufacturer**  
Fisher Scientific Company  
One Reagent Lane  
Fair Lawn, NJ 07410  
Tel: (201) 796-7100

**Emergency Telephone Number** For information **US** call: 001-800-ACROS-01 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

**WHMIS 2015 Classification** Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                                         |            |
|---------------------------------------------------------|------------|
| <b>Skin Corrosion/Irritation</b>                        | Category 2 |
| <b>Serious Eye Damage/Eye Irritation</b>                | Category 2 |
| <b>Specific target organ toxicity (single exposure)</b> | Category 3 |
| Target Organs - Respiratory system.                     |            |

#### Label Elements

**Signal Word**  
Warning

**Hazard Statements**  
Causes skin irritation  
Causes serious eye irritation  
May cause respiratory irritation

**Precautionary Statements****Prevention**

Avoid breathing dust/fume/gas/mist/vapors/spray  
Wash face, hands and any exposed skin thoroughly after handling  
Use only outdoors or in a well-ventilated area  
Wear protective gloves/protective clothing/eye protection/face protection

**Response**

IF ON SKIN: Wash with plenty of soap and water  
IF INHALED: Remove person to fresh air and keep comfortable for breathing  
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
Call a POISON CENTER/ doctor if you feel unwell  
Take off contaminated clothing

**Storage**

Store in a well-ventilated place. Keep container tightly closed  
Store locked up

**Disposal**

Dispose of contents/container to an approved waste disposal plant

### 3. Composition/Information on Ingredients

| Component                                                                             | CAS-No     | Weight % |
|---------------------------------------------------------------------------------------|------------|----------|
| L-Lysine,<br>N2-[(1,1-dimethylethoxy)carbonyl]-N6-[(9H-fluoren-9-ylmethoxy)carbonyl]- | 84624-27-1 | >95      |

### 4. First-aid measures

|                                        |                                                                                                                   |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                  | If symptoms persist, call a physician.                                                                            |
| <b>Eye Contact</b>                     | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.   |
| <b>Skin Contact</b>                    | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician. |
| <b>Inhalation</b>                      | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.      |
| <b>Ingestion</b>                       | Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.             |
| <b>Most important symptoms/effects</b> | None reasonably foreseeable.                                                                                      |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                             |

### 5. Fire-fighting measures

|                                     |                                            |
|-------------------------------------|--------------------------------------------|
| <b>Suitable Extinguishing Media</b> | Carbon dioxide (CO <sub>2</sub> ). Powder. |
|-------------------------------------|--------------------------------------------|

|                                         |                          |
|-----------------------------------------|--------------------------|
| <b>Unsuitable Extinguishing Media</b>   | No information available |
| <b>Flash Point</b>                      | Not applicable           |
| <b>Method -</b>                         | No information available |
| <b>Autoignition Temperature</b>         | No information available |
| <b>Explosion Limits</b>                 |                          |
| <b>Upper</b>                            | No data available        |
| <b>Lower</b>                            | No data available        |
| <b>Sensitivity to Mechanical Impact</b> | No information available |
| <b>Sensitivity to Static Discharge</b>  | No information available |

**Specific Hazards Arising from the Chemical**

Keep product and empty container away from heat and sources of ignition.

**Hazardous Combustion Products**

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Nitrogen oxides (NO<sub>x</sub>).

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**NFPA**

**Health**  
2

**Flammability**  
1

**Instability**  
1

**Physical hazards**  
N/A

## 6. Accidental release measures

|                                             |                                                                                                              |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Use personal protective equipment as required. Ensure adequate ventilation. Avoid dust formation.            |
| <b>Environmental Precautions</b>            | Should not be released into the environment. See Section 12 for additional Ecological Information.           |
| <b>Methods for Containment and Clean Up</b> | Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal. |

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                             |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid ingestion and inhalation. Avoid dust formation. Do not get in eyes, on skin, or on clothing.                                                         |
| <b>Storage.</b> | Keep container tightly closed in a dry and well-ventilated place.<br>To maintain product quality. Keep refrigerated. Store under an inert atmosphere. Protect from moisture. Incompatible Materials. Strong oxidizing agents. Strong acids. |

## 8. Exposure controls / personal protection

|                            |                                                                                                                                               |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Exposure Guidelines</b> | This product does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies. |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|

**Engineering Measures**

None under normal use conditions. Ensure that eyewash stations and safety showers are close to the workstation location.

**Personal protective equipment**

|                        |                                                                           |
|------------------------|---------------------------------------------------------------------------|
| <b>Eye Protection</b>  | Goggles                                                                   |
| <b>Hand Protection</b> | Wear appropriate protective gloves and clothing to prevent skin exposure. |

| Glove material | Breakthrough time | Glove thickness | Glove comments         |
|----------------|-------------------|-----------------|------------------------|
| Nitrile rubber | See manufacturers | -               | Splash protection only |
| Neoprene       | recommendations   |                 |                        |
| Natural rubber |                   |                 |                        |
| PVC            |                   |                 |                        |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

#### Respiratory Protection

No protective equipment is needed under normal use conditions.

**Recommended Filter type:** Particle filter Particulates filter conforming to EN 143

#### Environmental exposure controls

No information available.

#### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                                        |                                                               |
|----------------------------------------|---------------------------------------------------------------|
| Physical State                         | Solid                                                         |
| Appearance                             | White                                                         |
| Odor                                   | Odorless                                                      |
| Odor Threshold                         | No information available                                      |
| pH                                     | No information available                                      |
| Melting Point/Range                    | 135 - 138 °C / 275 - 280.4 °F                                 |
| Boiling Point/Range                    | No information available                                      |
| Flash Point                            | Not applicable                                                |
| Evaporation Rate                       | Not applicable                                                |
| Flammability (solid,gas)               | No information available                                      |
| Flammability or explosive limits       |                                                               |
| Upper                                  | No data available                                             |
| Lower                                  | No data available                                             |
| Vapor Pressure                         | No information available                                      |
| Vapor Density                          | Not applicable                                                |
| Specific Gravity                       | No information available                                      |
| Solubility                             | No information available                                      |
| Partition coefficient; n-octanol/water | No data available                                             |
| Autoignition Temperature               | No information available                                      |
| Decomposition Temperature              | No information available                                      |
| Viscosity                              | Not applicable                                                |
| Molecular Formula                      | C <sub>26</sub> H <sub>32</sub> N <sub>2</sub> O <sub>6</sub> |
| Molecular Weight                       | 468.55                                                        |
| Optical rotation                       | [α] <sub>D20</sub> = +3.4 deg, c=1, EtOAc                     |

## 10. Stability and reactivity

|                        |                                                                                           |
|------------------------|-------------------------------------------------------------------------------------------|
| Reactive Hazard        | None known, based on information available                                                |
| Stability              | Stable under recommended storage conditions. Hygroscopic.                                 |
| Conditions to Avoid    | Incompatible products. Excess heat. Exposure to moist air or water. Avoid dust formation. |
| Incompatible Materials | Strong oxidizing agents, Strong acids                                                     |

**Hazardous Decomposition Products** Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Nitrogen oxides (NO<sub>x</sub>)

**Hazardous Polymerization** Hazardous polymerization does not occur.

**Hazardous Reactions** None under normal processing.

## 11. Toxicological information

### Acute Toxicity

**Product Information** No acute toxicity information is available for this product

**Component Information**

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Irritation** No information available

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component                                                                             | CAS-No     | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|---------------------------------------------------------------------------------------|------------|------------|------------|------------|------------|------------|
| L-Lysine,<br>N2-[(1,1-dimethylethoxy)carbonyl]-N6-[(9H-fluoren-9-ylmethoxy)carbonyl]- | 84624-27-1 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** Respiratory system

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** No information available

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Do not empty into drains.

**Persistence and Degradability** No information available

**Bioaccumulation/ Accumulation** No information available.

**Mobility** No information available.

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a

hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

#### 14. Transport information

|                 |               |
|-----------------|---------------|
| <u>DOT</u>      | Not regulated |
| <u>TDG</u>      | Not regulated |
| <u>IATA</u>     | Not regulated |
| <u>IMDG/IMO</u> | Not regulated |

#### 15. Regulatory information

##### International Inventories

| Component                                                                          | CAS-No     | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS | ELINCS | NLP |
|------------------------------------------------------------------------------------|------------|-----|------|------|-----------------------------------------------|--------|--------|-----|
| L-Lysine, N2-[(1,1-dimethylethoxy)carbonyl]-N6-[(9H-fluoren-9-ylmethoxy)carbonyl]- | 84624-27-1 | -   | -    | -    | -                                             | -      | -      | -   |

| Component                                                                          | CAS-No     | IECSC | KECL | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|------------------------------------------------------------------------------------|------------|-------|------|------|------|------|------|-------|-------|
| L-Lysine, N2-[(1,1-dimethylethoxy)carbonyl]-N6-[(9H-fluoren-9-ylmethoxy)carbonyl]- | 84624-27-1 | -     | -    | -    | -    | X    | -    | -     | -     |

##### Legend:

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

##### Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

##### Other International Regulations

##### Authorisation/Restrictions according to EU REACH

##### Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component                                                                          | CAS-No     | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|------------------------------------------------------------------------------------|------------|----------------|------------------------------|---------------------------|--------------------------------------------|
| L-Lysine, N2-[(1,1-dimethylethoxy)carbonyl]-N6-[(9H-fluoren-9-ylmethoxy)carbonyl]- | 84624-27-1 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

| Component | CAS-No | Seveso III Directive (2012/18/EC) - | Seveso III Directive (2012/18/EC) - | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-----------|--------|-------------------------------------|-------------------------------------|----------------------------|------------------------------------|
|-----------|--------|-------------------------------------|-------------------------------------|----------------------------|------------------------------------|

|                                                                                               |            | Qualifying Quantities<br>for Major Accident<br>Notification | Qualifying Quantities<br>for Safety Report<br>Requirements |                |                |
|-----------------------------------------------------------------------------------------------|------------|-------------------------------------------------------------|------------------------------------------------------------|----------------|----------------|
| L-Lysine,<br>N2-[(1,1-dimethylethoxy)carbo<br>nyl]-N6-[(9H-fluoren-9-ylmetho<br>xy)carbonyl]- | 84624-27-1 | Not applicable                                              | Not applicable                                             | Not applicable | Not applicable |

## 16. Other information

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**Creation Date**

13-April-2011

**Revision Date**

28-December-2021

**Print Date**

28-December-2021

**Revision Summary**

This document has been updated to comply with the requirements of WHMIS 2015 to align with the Globally Harmonised System (GHS) for the Classification and Labelling of Chemicals.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**